

Course Code	405-04																																																						
Course Title	Incident Response and Digital Forensics																																																						
Credit	4																																																						
Course Category	Major Course																																																						
Level of Course	300-399 (Higher Level)																																																						
Teaching per week	4 Hrs (2 Hours Theory + 4 Hours Practical)																																																						
Minimum Weeks per Semester	15 Weeks (including Class work, examination, preparation, etc.)																																																						
Implementation Year	A.Y 2025-2026																																																						
Medium of Instruction	English																																																						
Purpose of Course	To equip students with the skills necessary to effectively respond to cybersecurity incidents and perform forensic investigations. The course emphasizes practical techniques for identifying, analyzing, and documenting cyber incidents using open-source forensic tools.																																																						
Course Objective	1) Understanding incident response methodologies. 2) Performing forensic analysis using open-source tools. 3) Identifying and mitigating security breaches. 4) Reporting forensic investigation findings.																																																						
Pre-requisites	Basic knowledge of cybersecurity and operating systems Familiarity with system and network concepts Prior exposure to Linux environment and command-line usage (preferred)																																																						
Course Outcome	CO1: Understand the phases of incident response and apply appropriate procedures. CO2: Perform forensic acquisition and analysis of digital evidence. CO3: Analyze malware behavior and identify Indicators of Compromise (IOCs). CO4: Investigate and interpret network traffic for suspicious activity. CO5: Create professional reports suitable for legal and regulatory compliance.																																																						
Mapping between Course Outcomes(CO) with Program Specific Outcomes(PSO)	<table border="1"> <thead> <tr> <th></th><th>PSO1</th><th>PSO2</th><th>PSO3</th><th>PSO4</th><th>PSO5</th><th>PSO6</th><th>PSO7</th><th>PSO8</th></tr> </thead> <tbody> <tr> <td>CO1</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr> <tr> <td>CO2</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr> <tr> <td>CO3</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr> <tr> <td>CO4</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr> <tr> <td>CO5</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr> </tbody> </table>		PSO1	PSO2	PSO3	PSO4	PSO5	PSO6	PSO7	PSO8	CO1									CO2									CO3									CO4									CO5								
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Course Content	Unit-1. Fundamentals of Incident Response 1.1. Introduction to Incident Handling 1.2. Phases of Incident Response (NIST Framework) 1.3. Identifying Indicators of Compromise (IoCs) 1.4. Incident Reporting and Documentation 1.5. Setting Up an Incident Response Lab Unit-2. Digital Forensic Techniques 2.1. Forensic Data Acquisition (Autopsy, FTK Imager) 2.2. File System and Memory Analysis 2.3. Network Traffic Analysis (Wireshark, Zeek) 2.4. Log Analysis and Threat Hunting 2.5. Email and Metadata Forensics Unit-3. Malware Analysis and Reverse Engineering 3.1. Identifying Malicious Software																																																						

	3.2. Static and Dynamic Malware Analysis (YARA, Cuckoo Sandbox) 3.3. Extracting IOCs from Malware Samples 3.4. Reverse Engineering Using Ghidra 3.5. Analyzing Ransomware Attacks Unit-4. Reporting and Legal Considerations 4.1. Documenting Forensic Findings 4.2. Preparing an Incident Report 4.3. Presenting Evidence in a Legal Context 4.4. Best Practices for Cyber Incident Management 4.5. Compliance and Regulatory Requirements (ISO, GDPR) Recommended Practical Exercises: • Collecting and Analyzing Digital Evidence • Performing Memory Analysis Using Volatility • Investigating Network Traffic with Wireshark • Conducting Malware Analysis with Cuckoo Sandbox • Writing a Comprehensive Digital Forensic Report
Reference Books	1. The Unofficial Guide to Ethical Hacking, Ankit Fadia, Macmillan Publishers India Ltd., 0333936795, 1st 2. Ethical Hacking 5-in-1 Book with Practicals,, Code Academy, 9788195765652, 2021 3. The Complete Ethical Hacking Book: A Comprehensive Beginner's Guide, , Orangebooks Publication, 935621204X, 1st 4. Cyber Safe Girl, Dr. Ananth Prabhu G, , 9789353820305, 5. An Ethical Guide to Hacking Mobile Phones, , Macmillan Publishers India Ltd., 1403928509, 6. Ethical Hacking: A Hands-on Introduction to Breaking In, Daniel G. Graham, No Starch Press, 9781718501874, 1st 7. Ethical Hacking: Techniques, Tools, and Countermeasures, , Jones & Bartlett Learning, 9781284248999, 4th 8. Hands-On Ethical Hacking and Network Defense, Michael T. Simpson, Nicholas Antill, Cengage Learning, 9780357509753, 4th 9. Certified Ethical Hacker (CEH) Version 12, EC-Council, EC-Council Academia, 9798885931052, 12th 10. Learn Ethical Hacking from Scratch, Zaid Sabih, Packt Publishing, 9781788622059, 1st 11. Incident Response & Computer Forensics by Jason Luttgens, Matthew Pepe, Kevin Mandia 12. "The Art of Memory Forensics" by Michael Hale Ligh et al. 13. "Malware Analyst's Cookbook and DVD" by Michael Hale Ligh et al 14. "Practical Malware Analysis" by Michael Sikorski and Andrew Honig 15. Official Documentation: Autopsy, Volatility, Wireshark, Cuckoo Sandbox, Ghidra
Teaching Methodology	Class Work, Discussion, Self-Study, Seminars and/or Assignments
Evaluation Method	50% Internal assessment. 50% External assessment.